

CALIBURN MISSION BAY

Technical Documentation

OverKey Technical Datasheet

Capability: OverKey Mesh VPN

Provider: Caliola Engineering

TRL: TRL 9

OVERVIEW

Zero-trust distributed mesh VPN enabling secure communications over untrusted networks in DDIL environments, replacing traditional hub-and-spoke VPN architecture.

KEY FEATURES

1. Zero-Trust Architecture
2. DDIL Operations
3. Mesh Networking
4. Cryptographic Segregation

TECHNICAL SPECIFICATIONS

Architecture: Zero-trust distributed mesh

Environment: DDIL (Disconnected, Degraded, Intermittent, Limited)

Encryption: End-to-end without central server processing

Transport: Mesh radios, 5G, Starlink, satellite

Bandwidth: Optimized for tactical edge (1200+ bps)

Protocols: NSA-approved IPsec, AES-256, ECDH/ECDSA over P-384

Vlans: Software-defined VLANs with crypto segregation

Management: Web-based topology control interface

Failover: No single point of failure

Rekeying: Immediate upon node blocklisting

Traffic: Multicast (1-to-many) and incast (many-to-1)